

Miguel López

Software Architect · Backend & Cloud

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PROFESSIONAL SUMMARY

Software architect with 7+ years of experience across backend, cloud and systems. I design modular, event-driven platforms, define domain boundaries and model REST APIs with Clean Architecture, DDD and CQRS in Python and TypeScript. I integrate messaging (Kafka, CDC), data and security (OIDC, policy-based authorization), build Vue/Nuxt frontends, and carry decisions into production with Docker, GitLab CI and observability. I combine architectural judgement, product thinking and hands-on delivery.

EXPERIENCE

Research & Development Engineer 2023 – present
ITI — Instituto Tecnológico de Informática · Valencia

- Design and evolution of a modular internal platform in Python (FastAPI), applying Clean Architecture, DDD and CQRS with dependency injection.
- Event-driven architecture with Kafka, the outbox pattern and CDC with Debezium; persistence in PostgreSQL, MongoDB, Redis and Elasticsearch.
- Security with OIDC/OAuth2 and policy-based authorization (Cerbos); integrations with Microsoft 365, Azure DevOps and custom Odoo modules (CRM, payroll, analytics).
- Vue 3 and Nuxt frontends in a monorepo (Turborepo, pnpm) with an in-house design system; testing with Vitest and Playwright.
- Maintenance of Flask APIs and legacy applications (Symfony, NestJS); CI/CD with GitLab and Docker, observability with Sentry and OpenTelemetry, and MCP servers for AI assistants.

FastAPI · DDD · CQRS · Kafka · PostgreSQL · Vue 3 · Nuxt · Odoo

Junior Software Developer / TSR 2020 – 2023
Webhelp · Google Project · Barcelona

- Design and support of internal applications and tools on Google Cloud and Google Workspace.
- Service architecture, data integration and Google APIs on GCloud and Apigee.
- Automation scripting, code analysis and bug reporting.
- Direct technical communication with the client.

GCP · Apigee · APIs · Bash

Web Developer 2020
SNIK Comunicaciones · Remote

- Technical design, development and maintenance of websites and web applications for a range of clients.

JavaScript · PHP · Web

IT Technician 2019 – 2020
Altura Town Hall · Altura

- User management in Active Directory and cloud infrastructure on AWS.
- Implementation of internal web tools: vacation management and ticketing system.

AWS · Active Directory · IT

Freelance Web Developer 2018 – 2019
Self-employed · Remote

- End-to-end architecture design, development and deployment of applications.
- UI in Figma/Photoshop, frontend in React/Vanilla, backend in FastAPI/Express and data in Mongo/SQL on GCP/AWS.

React · FastAPI · MongoDB · GCP

SELECTED PROJECTS

Modular event-driven platform
FastAPI services with Clean Architecture, DDD and CQRS. Kafka events, outbox and CDC with Debezium, policy-based authorization and polyglot persistence (PostgreSQL, Mongo, Redis, Elasticsearch).
FastAPI · DDD · CQRS · Kafka

Nuxt application monorepo
A suite of Nuxt apps in a Turborepo + pnpm monorepo with shared UI and testing packages, a design system shipped as Nuxt modules and Playwright E2E. Changes driven by specs with OpenSpec.
Nuxt · TypeScript · Turborepo · Playwright

MCP servers for AI assistants
Model Context Protocol servers in TypeScript on a shared kernel, connecting AI assistants to project-management and support tools.
MCP · TypeScript · Zod · AI

Personal portfolio — emelcd.github.io
Site in React + Vite + TypeScript with Tailwind and shadcn/ui; bilingual, with themes and accents. Open source.
React · Vite · TypeScript

TECH STACK

Backend Architecture
Python · FastAPI · Flask · SQLAlchemy · Pydantic · NestJS · Express · Symfony

Frontend
Vue 3 · Nuxt · TypeScript · Pinia · Vuetify · Tailwind · DevExtreme · Storybook

Data & Messaging
Kafka · Debezium · PostgreSQL · MongoDB · MySQL · Redis · Elasticsearch · RisingWave

Security & Integrations
OAuth2 / OIDC · JWT · Cerbos · LDAP · Microsoft Graph · Azure DevOps · Odoo

Cloud, DevOps & Observability
Linux · Docker · GitLab CI · Nginx · Ansible · GCP · AWS · Sentry · OpenTelemetry · Grafana

Quality & AI-assisted Development
Pytest · Vitest · Playwright · Jest · SonarQube · OpenSpec · MCP · Claude Code

METHODOLOGIES

Clean Architecture · DDD · CQRS · Event-Driven
Outbox + CDC · REST / OpenAPI · TDD · CI/CD
Spec-Driven Development · Agile / Scrum
Code Review

CERTIFICATIONS

Spec-Driven Development with AI (OpenSpec) 2026
ITI

Custom Module Development with Odoo 2025
ITI · taught by PUNT

Kubernetes: Deploying and Operating Containerized Applications 2025
ITI

Hexagonal Architecture and CQRS for Backend Systems 2024
ITI

Developing APIs with Google Cloud Apigee API Platform 2026
Google

Google IT Automation with Python Professional Certificate 2022
Google

EDUCATION

Higher Technician in Network Computer Systems Administration (ASIR) 2021
UNED · Buitrago

Continuous self-directed learning Ongoing
Official docs, open source and side projects

STRENGTHS

Technical communication · Problem solving
Autonomy · Teamwork · Continuous learning

LANGUAGES

Spanish Native
English Professional

INTERESTS

Open source · Automation · Self-hosting · Linux